qa-python-practice

Public repository | 14 commits | 4 merged pull requests | last commit August 2026

README.md

Practice project I built while returning to work. It contains small Python scripts and unit tests.

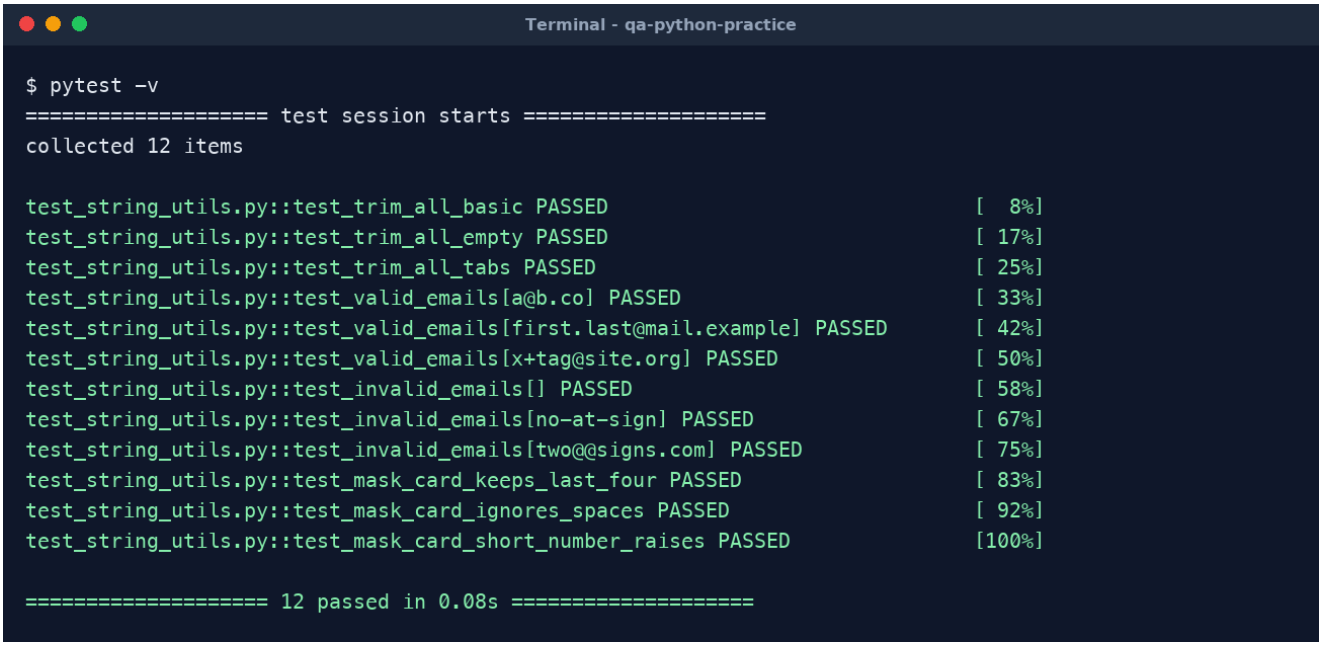
- string_utils.py: functions for trimming, validating email formats and masking card numbers.
- test_string_utils.py: 12 unit tests written with pytest, including parametrized tests for edge cases.
- csv_compare.py: script that compares expected and actual CSV test outputs and prints mismatched rows.

How to run: `pip install pytest`, then run `pytest -v`. Latest run: 12 passed in 0.08s.
Commit history shows feature branches merged through pull requests.

FILES

```
qa-python-practice/
  README.md
  string_utils.py
  test_string_utils.py
  csv_compare.py
```

TEST RUN



string_utils.py

```
"""Small string helpers used in my testing practice exercises."""
import re

EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[^@\s]+$")


def trim_all(value: str) -> str:
    """Trim the ends and collapse internal whitespace."""
    return " ".join(value.split())


def is_valid_email(value: str) -> bool:
    return bool(EMAIL_RE.match(value))


def mask_card(number: str) -> str:
    """Mask all but the last four digits of a card number."""
    digits = number.replace(" ", "")
    if len(digits) < 12:
        raise ValueError("card number too short")
    return "*" * (len(digits) - 4) + digits[-4:]
```

test_string_utils.py

```
import pytest
from string_utils import is_valid_email, mask_card, trim_all


def test_trim_all_basic():
    assert trim_all("  hello   world ") == "hello world"


def test_trim_all_empty():
    assert trim_all("   ") == ""


def test_trim_all_tabs():
    assert trim_all("a\t\tb") == "a b"


@pytest.mark.parametrize("email", ["a@b.co", "first.last@mail.example", "x+tag@site.org"])
def test_valid_emails(email):
    assert is_valid_email(email)


@pytest.mark.parametrize("email", ["", "no-at-sign", "two@@signs.com"])
def test_invalid_emails(email):
    assert not is_valid_email(email)


def test_mask_card_keeps_last_four():
    assert mask_card("4111111111111111") == "************1111"


def test_mask_card_ignores_spaces():
    assert mask_card("4111 1111 1111 1111") == "************1111"


def test_mask_card_short_number_raises():
    with pytest.raises(ValueError):
        mask_card("1234")
```

csv_compare.py

```
"""Compare expected and actual CSV outputs and print the mismatched rows."""
import csv
import sys


def load(path):
    with open(path, newline="") as fh:
        return list(csv.reader(fh))


def compare(expected_path, actual_path):
    expected, actual = load(expected_path), load(actual_path)
    mismatches = []
    for i in range(max(len(expected), len(actual))):
        e = expected[i] if i < len(expected) else None
        a = actual[i] if i < len(actual) else None
        if e != a:
            mismatches.append((i + 1, e, a))
    return mismatches


if __name__ == "__main__":
    for row, e, a in compare(sys.argv[1], sys.argv[2]):
        print(f"row {row}: expected={e} actual={a}")
```

COMMIT HISTORY

Date	Message
2026-08-21	Merge PR #4: chore/readme
2026-08-12	Update README with run instructions
2026-08-04	Add tab and empty-string tests
2026-07-26	Merge PR #3: feature/csv-compare
2026-07-19	Add csv_compare script
2026-07-11	Merge PR #2: feature/card-masking
2026-07-05	Test mask_card edge cases
2026-07-03	Add mask_card helper
2026-06-28	Merge PR #1: feature/email-validation
2026-06-24	Parametrize email tests
2026-06-20	Add pytest and first tests
2026-06-17	Add email validation
2026-06-15	Add trim_all helper
2026-06-14	Initial commit: README and project layout